

3rd International Conference on Industry 4.0 and Smart Manufacturing

Key Performance Indicators and Industry 4.0 – A structured approach for monitoring the implementation of digital technologies

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Abstract

Concerns regarding digital transformation have forced manufacturing organizations to rethink their operations and processes to gain a competitive advantage. For several companies, the design and implementation of indicators are deemed crucial for better management and control of emerging concepts and ideas regarding digital technologies. Despite the quick shift towards digital transformation has captured the attention of both the industry and academia, very few efforts have been made to streamline useful indicators able to monitor the implementation of Industry 4.0 (I4.0) technologies. Owing to these considerations, this paper attempts (i) to design and implement several Key Performance Indicators able to monitor the evolution of core processes that will benefit from the I4.0 technologies, and (ii) to provide an easy-to-use approach that can be implemented in practice to guide decision-makers in the diffusion, adoption, and performance evaluation of I4.0 technologies. The effectiveness and practicality of the methodology are illustrated using a case study concerning an industrial project implemented in a leading international company in the power generation industry.

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Peer-review under responsibility of the scientific committee of the 3rd International Conference on Industry 4.0 and Smart Manufacturing

Keywords: Industry 4.0; Indicators; KPI; Digital technologies; Digital transformation; Case study

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1. Introduction

The potentials offered by the increasing application of digitization are reshaping the competitive dispositions of organizations, their customer and employee interrelation, and market positioning [1]. Hence, the capability of firms to grasp the concepts and applications behind digitization has become crucial in gaining a competitive advantage [2].

The area of digital transformation contains a dimension that has been capturing the interest of academics and practitioners concerning the prospects and the effect of applying digitisation to organizations, which is commonly termed Industry 4.0 (I4.0). Rapid changes brought by I4.0 modified how an organization operates. The principle of I4.0 is the interconnectivity of digital technologies, devices, and processes, which enables the operation of autonomous manufacturing models, able to perform in a decentralised decision setting with minimal human interference, and capable of connectedly working together along the stages of the production process and across several stages of the supply chain [1]. Furthermore, companies need to create appropriate strategies to aid their preparation for future industrial developments.

For most companies, the design and implementation of indicators are deemed crucial for better management and control of emerging concepts and ideas regarding digital technologies. Furthermore, indicators are significant for an appropriate and efficient allocation of resources, and assessment of performance at a specific innovation stage. In policymaking initiatives, having an appropriate set of indicators can aid an organization's status and level of the I4.0 implementation, further recognizing the relevance and suitability of innovation activities completed to realize the full implementation of I4.0.

Despite the quick shift towards digital transformation has captured the attention of both the industry and academia, very few efforts have been made to streamline useful indicators able to monitor the implementation of I4.0 technologies (Section 3).

This paper thus attempts (i) to design and implement several Key Performance Indicators (KPIs) able to monitor the evolution of core processes that will benefit from the I4.0 technologies, and (ii) to guide decision-makers in I4.0 technologies diffusion and adoption as well as performance evaluation.

Note that we do not attempt to develop mathematical models or algorithms to build KPIs and capture their interrelationships. Rather, we intend to provide an easy-to-use approach that can be implemented in practice to guide decision-makers in the diffusion, adoption, and performance evaluation of production lines, departments, or machines influenced by I4.0 technologies. Regarding this, an industrial case study is considered to show the potentialities of the proposed method. In particular, the effectiveness and practicality of the methodology are illustrated using an industrial project implemented in a leading international company in the power generation industry.

The remaining part of this paper is organized as follows. Section 2 outlines the methodological steps undertaken. In Section 3, a relevant literature review that analyses indicators able to monitor the evolution of core processes that will benefit from the I4.0 technologies is provided. In order to show the operating principles and results of the proposed approach, a practical application in an industrial context is presented in Section 4. Section 5 discusses the proposed approach with its strengths and limitations. The conclusions and future possible developments are drawn in Section 6.

2. Methodological steps

In order to achieve the above-mentioned goals, we conducted our research according to four methodological steps that are listed below.

Step 1. Literature Review. First, a literature review was conducted to investigate how the digital transformation has captured the attention of both the industry and academia. Particularly, we focused on indicators able to monitor the evolution of core processes that will benefit from the I4.0 technologies.

Step 2. Case study. In this section, an easy-to-use approach that can be implemented in practice to guide decision-makers in the adoption, diffusion, and performance evaluation of I4.0 technologies is proposed. Specifically, the application deals with a multinational company in the power generation industry. The case study permits us to further discuss some operational issues of the proposed approach, as well as to testify its practical applicability, effectiveness, and potential beyond the considered industry sector.

Step 3. *Discussion*. The strengths and limitations of the approach are discussed also to ensure its future application to further industrial contexts.

Step 4. *Conclusions and future research*. The last step concludes and suggests the future needs of research to promote further improvements and development of the proposed approach.

3. Literature Review

In order to analyse studies in the field of KPIs able to monitor the evolution of core processes that will benefit from the I4.0 technologies, a review using a keyword search in Scopus supplemented by a subsequent forward and backward search was conducted. The following search string in Scopus was used: (TITLE-ABS-KEY (indicator) AND TITLE-ABS-KEY (industry 4.0) AND TITLE-ABS-KEY (digital AND transformation)) AND (LIMIT-TO (SUBJAREA, "ENGI")) AND (LIMIT-TO (LANGUAGE, "English")). From the 15 identified works (retrieved on 1st September 2021), 11 have been rated to be relevant for this research. They were selected after the first screening of the title and abstract, and by using a subsequent forward and backward search approach.

What emerges from the analysis is that only a few efforts have been made to streamline useful indicators able to monitor the implementation of I4.0 technologies. Himang et al. (2020) [3] established an indicator system for adopting I4.0 within the context of the three stages of innovation adoption. They identified a set of 63 indicators and presented a case study from two manufacturing firms. However, they do not identified the areas of the firms that were benefiting from I4.0. technologies. As a result, the indicators give limited indications of where to intervene in the processes in cases of bad performance. In addition, the identification of the metrics for each indicator proposed was out of the scope of their work.

Some analysed papers focus on the implementation of a single I4.0 technology like Cyber-physical system [4], cloud computing adoption [5], adoption of the Internet of Things [6], and smart manufacturing [7]. Thus, it is very difficult to identify the impact of low-performance indicators to the overall process.

Furthermore, additional indicators are limited to a specific context like the deep-drawing process [8], agroindustry [9], or energy management in manufacturing [10] giving a restricted indication of causes for inefficiencies within I4.0.

Finally, several works present metrics at a conceptual and qualitative level without an empirical validation in practice. Specifically, Ocampo et al. (2021) [11] attempted to extract I4.0 indicators from relevant literature and integrate these indicators in strategy formulation by providing a categorization system according to technological, organizational, inter-organizational, and social, and regulatory clusters. Zizek et al. (2020) [12] outlined the role and importance of KPIs in the frame of I4.0 implementation, and Alekseev et al. (2019) [13] studied the main stages of formation of I4.0 and determine the KPIs of its development at each of them. Finally, de Souza et al. (2020) [9] performed a systematic review of the literature on economic, social, and sustainable pillar indicators in agroindustry companies and I4.0.

Based on the state of the art analysis, and to the best of the author's knowledge, a structured and operative approach able to (i) design and implement several KPIs able to monitor the evolution of core processes that will benefit from the I4.0 technologies, and (ii) to provide an easy-to-use approach that can be implemented in practice to guide decision-makers in the diffusion, adoption, and performance evaluation of I4.0 technologies actions is still missing.

4. Case study

In this section, a real case study is presented to show the practical applicability of the proposed methodology. The application deals with a multinational company in the power generation industry, to which it brings an integrated model embracing turnkey power plants construction, power equipment (gas & steam turbines, generators, and microturbines), manufacturing, and services. In particular, the approach proposed in this paper has been recognized by the company's top management as a valid method to design and implement several KPIs able to monitor the evolution of core processes that will benefit from I4.0 technologies.

The digital transformation of the company began years ago and covers all key processes in the value chain, from marketing and sales through to product development that is increasingly integrated at a digital level with service

activities, as well as embracing production, supply chain management, and the erection and startup of machines and plants. Human capital continues to be the focus of the digital transformation, which addresses the technology to support the ongoing improvement of safety at work, the skills and quality of resources, and the overall organization of the company.

4.1. A brief overview of the industrial project

The industrial project was conducted within a workgroup led by a responsible production manager. The team was composed of academic personnel and company experts in different areas such as design, manufacturing, material handling, quality, maintenance, and logistic suppliers. The group has worked for three months, mainly through roundtable discussions.

The project has been organized in eight Application Areas (AAs) of particular interest for the company, each representing a goal to achieve (Fig. 1):

- AA1. Production Operations Management
- AA2. Physical Operations Management
- AA3. Production Asset Performance Management
- AA4. Technical Data Management
- AA5. Quality Data and Additive Manufacturing Management
- AA6. Smart Safety
- AA7. Smart Training
- AA8. Cyber Security

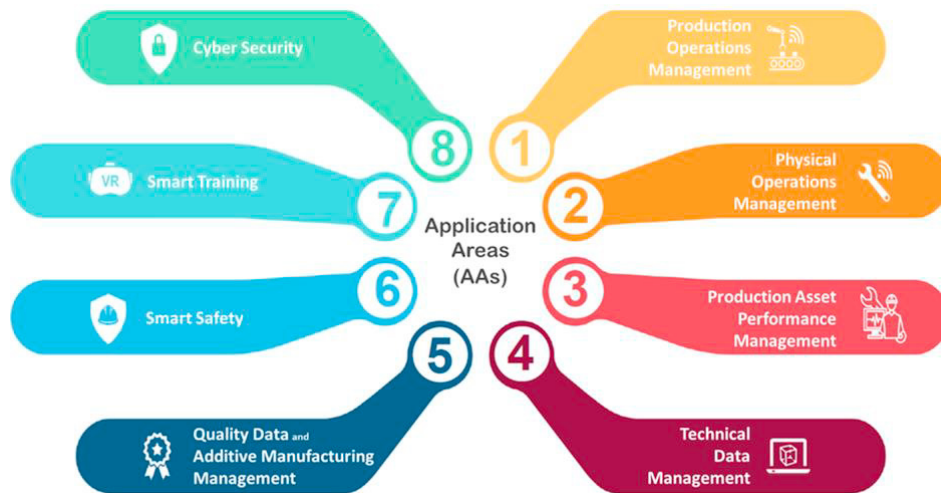


Figure 1. Application Areas of the industrial project.

Specifically, the project addresses the development and application of the technologies of I4.0 based on a model organized in three main strands of action:

1. The first three AAs (AA1. Production Operations Management, AA2. Physical Operations Management, and AA3. Production Assets Performance Management) make it possible to extend digitalization to all production lines, improving their efficiency and quality, as well as introducing innovations such as the Smart Tracking of all production factors and Predictive Maintenance achieved by retrofitting in-service plants and machineries.
2. The next two AAs (AA4. Technical Data Management and AA5. Quality Data and Additive Manufacturing Management) capitalize on the use of new big data technologies in the technical and quality areas to facilitate the

collection, analysis, and use of data to support the decision-making and actions of factory resources. Also promoted is the use of Additive Manufacturing (3D Printing) in production lines.

3. The remaining three AAs (AA6. Smart Safety, AA7. Smart Training, and AA8. Cyber Security) refer to innovative digital processes and applications (e.g., augmented and virtual reality, simulation, simplified man-machine interface, etc.) that, after development and testing in the factory, could be replicated and adapted for other operating scenarios, including worksites for new plants and service work in the field (maintenance and assistance).

Table 1 reports the identification of the eight AAs, and their decomposition in operational activities.

Table 1. Identification of the eight Applications Areas, and their decomposition in operational activities.

ID	Application Area	Operational Activity
AA1	Production Operations Management	Manufacturing Information Dispatching
		Product Definition and Routing Management
		Product Operations Tracking
		Digital Twin
		ERP Integration
		Planning and Scheduling
AA2	Physical Operations Management	Physical and Logical Tracking
		Layout Modelling in the Factory Environment
AA3	Production Asset Performance Management	Production Asset Monitoring
		Predictive Maintenance
		Enhanced CMMS
AA4	Technical Data Management	Electronic Work Instruction
		Machine Control Programs
AA5	Quality Data and Additive Manufacturing Management	Quality Data Collection and Management
		Statistical Analysis of Quality Data
		Part Components Repair
AA6	Smart Safety	PPE Management
		Detection and Surveillance of High Areas Risk
AA7	Smart Training	Virtual Reality
		Development of Man-Machine Interface
AA8	Cyber Security	Cyber Security Enforcement
		Monitoring and Certification of Production Cycle

4.2. The approach and the logic behind the KPIs

In this section, an easy-to-use approach that can be implemented in practice to guide decision-makers in the adoption, diffusion, and performance evaluation of I4.0 technologies is proposed. Here KPIs are presented concerning three different categories:

- *Diffusion-KPIs*. This category of indicators allows evaluating the diffusion of technologies, both hardware and software, related to I4.0 and relevant for the AA
- *Adoption-KPIs*. This category collects KPIs which investigate the adoption of an I4.0 technology relevant to a specific AA
- *Performance-KPIs*. This set of KPIs aims to measure and monitor the evolution of operational performance across different AAs in conjunction with the introduction of new technologies. A general measure of Performance-KPI is given by the ratio between corresponding elements, both expressed in the same unit of measurement

According to this classification, several different KPIs were proposed and an original sample of 90 KPIs was collected. The original sample has been conceived with coverage of journal articles [3]-[8], conference proceedings [14]-[15], and an international standard [16]. KPIs were compared, and all inconsistencies were resolved through discussion between the academician personnel and company experts until consensus was reached. After the exclusion of duplicates, the sample was reduced from 100 to 90. Finally, 23 KPIs were excluded because considered extra-scope. Indicators were deemed relevant if they satisfy the goal of assessing the level of the I4.0 implementation to measure and monitor operational performances across the AAs of the company. Accordingly, the final sample contained 57 KPIs.

The indicators, in order to be practical, were specific and structured, meaning that their definition should comprise, highlight, and isolate the key factors and then suggest countermeasures. Moreover, each indicator (when possible) was part of a complementary and formally similar KPI, to provide the management with a consistent and easy-to-use tool to control and adjust the all-around system performance.

Specifically, KPIs were inspired by the tabular structure reported in the ISO 22400 [15], which provides a deep overview of the concepts, the terminology, and the methods to describe KPIs to manage manufacturing operations. Consequently, each KPI reported the following descriptive elements:

- ID. A user-defined unique identification of the KPI
- Type. Diffusion, Adoption, or Performance KPI
- Name. Name of the KPI
- Description. A brief description of the KPI
- Formula. The mathematical formula of the KPI specified in terms of elements
- Trend-Target. Is the information about the improvement direction, higher is better or lower is better
- Benefit. Identification of the benefit that the KPI is relevant for, which can be, for instance, time reduction, cost reduction, quality increase, automatic tracking
- Timing. A KPI can be calculated either in real-time – after each new data acquisition event, on-demand – after a specific data selection request, periodically – done at a certain interval, e.g. once per day, twice a week
- Notes. This field allows noting elements of attention

In order to better understand the logic behind the KPIs, Table 2, Table 3, and Table 4 report an example of three different KPIs, respectively of Diffusion, Adoption, and Performance involved in the calculation of the Overall Equipment Effectiveness (OEE). The structure of the three tables identifies KPI elements in the left columns and describes each element in the right column. Most of the information required to build the KPIs could be directly obtained from commonly available data within companies' Information Systems. In the majority of cases, these data require very simple manipulation. However, where data appear to be incomplete, the lack could be recovered by implementing proper recording procedures (e.g. simple papers, electronic spreadsheets, or more complex software applications).

In order to support the operational stage of our methodology, we proposed a matrix where each KPI category is reported in the rows, while the AAs, and their decomposition in operation activities, are shown in the columns (Fig. 2). Each element of the matrix reports the ID of a specific KPI. Note that a three-color representation is used: orange means the KPI is under design, yellow it is being validated, green it is implemented. If a KPI does not arise in a given location, then the corresponding cell in the matrix will be empty.

The matrix also allows to identify the production line (PL), shop floor (SF), or machine (MA) where the KPI is designed, validated, or implemented. If the same KPI is used to monitor the same AAs – i.e. operational activities, in different parts of the factory, the columns are divided into several sub-columns equal to the implementation locations.

Hence, the matrix can be used as a very effective operative and simple tool able to visualize the large amount of KPIs collected during the analysis in a structured and immediate way. In this way, it is possible (i) to monitor the status of a KPI, (ii) to observe the production lines, departments, or machines that are benefiting from Industry 4.0 technologies, and (iii) to guide decision-makers in the diffusion, adoption, and performance evaluation of production lines, departments, or machines influenced by I4.0 technologies.

Table 2. An example of Diffusion-KPI.

ID	1
Type	Diffusion-KPI
Name	Diffusion of HW devices to implement digital data collection
Description	The indicator measures number of work machines equipped with HW devices necessary to implement digital data collection compared to the total number of monitored machines
Formula	$\frac{\text{Number of machines equipped with HD devices to implement digital data collection}}{\text{Number of monitored machines}}$
Trend-Target	100%
Benefit	Effective and efficient use of HW devices to implement digital data collection
Timing	Once a week
Notes	

Table 3. An example of Adoption-KPI.

ID	2
Type	Adoption-KPI
Name	Adoption of digital sheets to calculate the OEE-Availability metric
Description	The indicator evaluates the number of OEE sheets received in digital format compared to the total number of digital cards that the machine can dispense*
Formula	$\frac{\text{Number of OEE sheets received in digital format}}{\text{Number of OEE sheets potentially receivable in digital format}}$
Trend-Target	100%
Benefit	- Reduction/Elimination of the collection and delivery of paper sheets - Reduction/Elimination of printing activities
Timing	Once a week
Notes	*The personnel must be trained to properly use the new digital sheets

Table 4. An example of Performance-KPI.

ID	3
Type	Performance-KPI
Name	Variation in data collection times
Description	The indicator allows to evaluate the mean time required to carry out data collection activities compared to a reference time*
Formula	$\frac{\text{Average time of data collection after the introduction of digital format}}{\text{Average time of data collection before the introduction of digital format}}$
Trend-Target	The lower, the better
Benefit	- Reduction of data collection times - Availability of data in real-time, which allow implementing countermeasures in a timely and reliable manner
Timing	Once a week
Notes	*Reference time is the average time calculated in the period before the introduction of the digital format for data collection

Application Area (ID)	Production Operations Management (AA1)					Physical Operations Management (AA2)			Production Asset Performance Management (AA3)			Technical Data Management (AA4)		Quality Data and Additive Manufacturing Management (AA5)			Smart Safety (AA6)		Smart Training (AA7)		Cyber Security (AA8)	
	Manufacturing Information Dispatching	Product Definition and Routing Management	Production Operations Tracking	Digital Twin	ERP Integration	Planning and Scheduling	Physical and Logical Tracking	Layout Modelling in the Factory Environment	Production Asset Monitoring	Predictive Maintenance	Enhanced CMMS	Electronic Work Instruction	Machine Control Programs	Quality Data Collection and Management	Statistical Analysis of Quality Data	Part Components Repair	PPE Management	Detection and Surveillance of High Areas Risk	Virtual Reality	Development of Man-Machine Interface	Cyber Security Enforcement	Monitoring and Certification of Production Cycle
Operational Activity	MA01	MA01	MA01	MA05	MA05	MA01	MA01	MA01	MA01	MA04	MA05	MA01	MA05	MA01	MA01	MA05	SL02	SL02	PI01	PI01	PI01	PI01
	1	51	51				6		5					5		33	39	39	54	55	56	57
Diffusion-KPI	43		52				24		12					13		35						
	2						25															
Adoption-KPI	4											26	30	2								
	45																					
Performance-KPI	3	10	53	11	36	36	7		53	14				53	15		17	17				
	47					40	7			14					16		18	18				
							8		19	19	19				16		25	25				
							8		20	20	20				31		34	34				
							9		21	21	21				32		37	37				
							9		22	22	22				38							
							29		23	23	23				41							
							48		24	24	27				44							
							49		28	28	28				46							
							50		42	42	42											

Figure 2. The status of KPIs in the Application Areas.

5. Discussion

Before concluding the paper, some final remarks concerning our methodology, along with practical implications for its implementation, deserve to be made. Despite the quick shift towards digital transformation has captured the attention of both the industry and academia, as deeply discussed in Section 3, very few efforts have been made to streamline useful indicators able to monitor the implementation of I4.0 technologies.

As a response, in this paper, we present a set of KPIs able (i) to design and implement several KPIs able to monitor the evolution of core processes that will benefit from the I4.0 technologies, and (ii) to provide an easy-to-use approach that can be implemented in practice to guide decision-makers in the diffusion, adoption, and performance evaluation of I4.0 technologies.

In order to support the operational stage of our methodology, we proposed a matrix able to visualize the KPIs collected during the analysis. The matrix permits to highlight, in a structured and immediate way, the status of a KPI, observing the areas of the factory that are benefiting from I4.0. technologies.

We considered the case of a multinational company in the power generation industry. Specifically, we identified eight AAs of particular interest for the company, considering a “quite low” level of detail to present the methodology in the most general context possible, without focusing on a specific industry sector nor a particular production setting (Make-to-Order, Make-to-Stock, or other). This choice does not jeopardise the possibility to apply the methodology in a more specific environment.

6. Conclusions and future research

According to the results obtained in the case study, the company's top management is now evaluating the opportunity to extend this methodology to the small and medium enterprises forming part of the supply chain that supports the company's competitiveness on national and international markets.

As a future study, the authors believe that is widely demonstrated that the methodology, due to its operative and structured approach, has the potential to be effectively applied in many industrial sectors after appropriate and specific modification and adaptation, through the study of different KPIs and the selection of different application areas. In addition, it could be useful to study the interaction between different KPIs to evaluate how the implementation of a countermeasure affects different parts of the system under analysis.

Acknowledgments

The authors would like to thank Consorzio Quinn (<https://www.consorzioquinn.it/>) for allowing us to take part in the project analysed in this study, without which this research paper would not be possible.

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